ABHINAV

ML Engineer | AI / IoT Developer | XAI · RAG · QML

Building Intelligent Systems · Explainable Models · Edge AI · Cloud-Connected IoT

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LinkedIn | GitHub | SnifTern | ParkIn

PROFESSIONAL SUMMARY

B.Tech CSE (AI/ML) student at Bengal Institute of Technology with expertise in Machine Learning, Explainable AI (XAI), Retrieval-Augmented Generation (RAG), and Quantum Machine Learning (QML). Built and deployed full-stack AI applications, real IoT systems on AWS IoT Core & ESP32, and production NLP fraud-detection platforms. Managing Lead of PredAIction — BIT's official AI & ML club. Intermediate across the complete ML pipeline from data preprocessing and model training to deployment and explainability.

EDUCATION

B.Tech – CSE (Artificial Intelligence & Machine Learning) *Bengal Institute of Technology, Kolkata* *Aug 2023 – Jun 2027*

- Specialization in AI & ML | Active Contributor to research, Tech fests & AI community

EXPERIENCE

Managing Lead *PredAIction – Official AI & ML Club, BIT* *Sep 2025 – Present*

- Leading 100+ member AI & ML community; running workshops, hackathons & industry speaker sessions
- Mentoring juniors in ML, deep learning, Python, data science; driving collaborative research projects

PROJECTS

ParkIn – AI-Powered Smart Parking System *Python · Flask · Scikit-learn · Supabase · React Native* *2025 – Present*

1st Place – GenFusion AI Competition, Tech Fest BIT Live GitHub Reop

- Led 4-member team to build cross-platform smart parking platform with real-time search, booking & space monetization
- AI recommendation engine (Flask API), demand_forecaster.py & price_predictor.py ML modules; Supabase backend, React Native UI — designed for future IoT sensor & ANPR integration

SnifTern.ai – Internship Fraud Detection Platform *Python · Flask · Scikit-learn · NLP · OCR · ReportLab* *2025 – Present*

1st Place – Envision Software Competition, Tech Fest BIT Live GitHub Repo

- ML platform detecting fake internship postings via Logistic Regression + TF-IDF; AI modules for salary anomaly, interview risk scoring & description quality assessment
- Multi-platform scraping (LinkedIn/Indeed/Glassdoor), OCR for screenshots, PDF export, multi-language (EN/HI/BN); 0–100 fraud scoring with company legitimacy database — deployed live

NAFLD_XAI – Liver Disease Detection with Explainable AI GitHub Repo *Python · XGBoost · SHAP · LIME · Pandas · Streamlit* *2026 – Present*

- XGBoost pipeline for NAFLD staging from clinical data; SMOTE for class imbalance — 80%+ accuracy; SHAP & LIME explainability

RAG Pipeline – Domain-Specific Document Q&A *Python · LangChain · HuggingFace Embeddings · Groq API* *2026 – Present*

- FAISS vector store + semantic chunking + HuggingFace embeddings for PDF Q&A; grounded context injection significantly reduces LLM hallucinations; multi-turn conversation support

Quantum ML Research – VQC Classification Experiments *Python · Qiskit · Scikit-learn* *2026 – Present*

- Exploring Variational Quantum Circuits for classification; benchmarking quantum vs. classical ML; investigating feature maps, entanglement strategies & shot-noise effects

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, Java
ML / AI: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Hugging Face Transformers
XAI / RAG / QML: SHAP, LIME, LangChain, RAG Pipelines, Qiskit
CV & Vision: OpenCV
Web / Full Stack: Flask, FastAPI, React.js, Node.js, HTML/CSS, Flutter
Databases: MongoDB, Supabase
IoT & Cloud: ESP32/Arduino, MQTT, AWS IoT Core, AWS IoT Device Management, Docker, Kubernetes
Tools: Git, GitHub, Colab, Streamlit, Antigravity, VS Code, Linux, Figma, Canva

CERTIFICATIONS

AWS IoT Foundations · Managing AWS IoT Devices at Scale · Handling AWS IoT Device Data & States (Apr 2026) | Data Structures Using Python – Udemy (May 2025) | Intro to LLMs · Prompt Design in Vertex AI · Responsible AI – Google | Crash Course in AI & ML – Great Learning (Nov 2024) | Google Cloud Data Analytics · Full Stack Java · Python for Beginners – Simplilearn

ACHIEVEMENTS

1st – GenFusion (AI) · 1st – Envision (Software) · 2nd – Electronovation (Hardware IoT) — Team Lead all three, Tech Fest BiT | Runner-up – College Hackathon | 2nd – Video Making Competition | Smart India Hackathon (SIH) Participant | Build with India Hackathon | Snapdragon Insider

LANGUAGES

English (Professional) | Hindi (Native)